2.2 Exercise 2 Data collection campaign: building a smell cartography

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Smart Analytics for Big Data

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Request

- Build a table with **four columns:**
    1. **Type of data to collect:** specify what kind of information will be gathered for the study
    2. **Associated "V's":** indicate which Big Data characteristics (Volume, Velocity, Variety, Veracity, Value) describe each data type
    3. **Contribution to smell identity:** discuss how each data type helps build the smell identity of a street, an area, a quarter, and the whole city
    4. **Strategy of data collection:** specify one or several methods or sources for gathering the data

Answer

Table 1 at the end of the document presents the four requested columns, followed by a brief explanation of the different motivations and reasoning behind their creation.

Smell: This specific data was chosen for obvious reasons given the assignment. Initially, there was an uncertainty about whether to provide participants with a predefined list of smells to choose from. However, after reading [1], I found that most studies on urban scents and smells allow participants to freely describe their perceptions in their own words. The limit of three words was imposed to ensure concise descriptions while still preserving user freedom of choice. This approach also facilitates later grouping of similar words or meanings. It is important to note that using a fixed set of smells could simplify the analysis but may introduce bias due to the selection of those predefined smells, so allowing a free-text descriptions encourages more honest and authentic responses. A predefined list could still be useful if a previous study on the smells of the Grenoble area existed, as it could help verify whether prior groupings and summaries accurately represent reality.
For the collection of these data, various methods can be employed. Existing maps or apps, such as SmellMyCity [2], can help gather data, allowing users to describe the smells they

perceive along with their precise location. As explained in [1], smells can be collected in multiple ways, including:

- Questionnaires or interviews conducted before, during, and/or after the walk
- Using maps, photography, audio notes, or digital tools to document where and what smells were detected
- Employing electronic noses to objectively measure smell intensity, sometimes in combination with human observations in hybrid smellwalks
- Classifying them into categories such as food, nature, traffic, or waste

Limiting other senses, such as sight or sound, can help participants focus solely on the smells they perceive without being influenced by their surroundings.

Location: Specifying the location of the collected data and associated smells is crucial for mapping and creating a graphical representation of urban smellscapes. Street surveys are one common approach: in some studies, volunteers were positioned along predefined paths and asked to describe the smells they encountered, while in other cases, they were free to move around the city.
The method of data collection can vary depending on the desired level of detail. The previously mentioned apps [2] can be used to identify smells based on users' exact locations. For a more precise description of specific streets, exact geographic coordinates or street and civic numbers should be recorded. For a broader overview, the city can be divided into larger zones, with perceived smells assigned to these areas [1].
This choice also affects the Big Data characteristics of the dataset. Collecting precise coordinates increases the *Variety* of data and may require higher *Velocity* to manage frequent updates or dense observations. Aggregating data into larger zones reduces spatial resolution and variety, but simplifies data handling and analysis, providing a more general yet still informative urban smell profile. Lcation data can generate a large amount of information, especially if collected at high spatial resolution (e.g., exact GPS coordinates for many participants over time). Each observation adds to the dataset, and when combined across the whole city, the total number of records can grow quickly. Managing and storing this large *Volume* of spatial data is an important consideration for urban smell mapping projects.

Sensation: This data is useful because it adds an emotional and qualitative dimension, helping to understand livability and comfort perception. It can also be valuable for clustering and correlation analyses. Campenni, Crawley, and Meier [3] showed that the same lavender smell can affect people differently depending on how it's described: if it was called "stimulating," heart rate went up, but if it was called "relaxing," heart rate went down. Adding a simple label like "pleasant" or "unpleasant" can help the study capture how different people perceive the same smells and how the location of a smell can influence that perception. This also ties into the Big Data aspects of *Veracity* and *Variability*: people's smell reports are subjective, so the reliability isn't perfect, and perceptions change depending on the individual and the time. It's important to keep these in mind when collecting or analyzing smell data.

To collect this type of data, volunteers could simply be asked to indicate which of the three categories best described what they were smelling. This information can be used to group smells or identify more pleasant areas of the city.

Meteorological conditions: It means collecting the information at the time and location of the smell surveys, such as temperature, wind, rain, and humidity. These conditions are usually obtained from meteorological records corresponding to the exact time and place of the observation.
Including meteorological data is important because weather can influence how smells are perceived. From a Big Data perspective, this data affects *Variability*, since weather conditions change over time and space, and *Velocity*, because frequent updates or high-resolution temporal data may be needed to match the smell observations accurately. So collecting these data can help to interpret temporal and environmental patterns in urban smellscapes. It's also important to notice how this category contains different type of data and collecting them for all the smell observation can create a huge *Volume* of data that must be stored and manege correctly.

Time: Knowing the time of observation is important because smell perception changes throughout the day. Carskadon et al. [4] found that people are most sensitive to smells in the late afternoon and least sensitive in the early morning, so the time of day really matters when assessing odors. In city smell mapping it's also important to understand how different areas can have different smells at different times. For example, a neighborhood might smell like bread or food during the day when bakeries are open, but in the evening or on weekends, it might smell more like alcohol due to bars and pubs. This can help us also understand how the smell of the cities change during the day, so to maybe be able to create a map of smell for the day and nigth time.
From a Data Analysis perspective, this affects *Velocity*, because smells can change quickly and need to be recorded in real time, and *Variability*, since both temporal and spatial differences influence perception. Collecting this data can be as simple as keepinf track of the time when a participant reports a smell, or automatically through an app that records timestamps along with the smell observation.

Human activities in the zone: Tracking human activities, such as the presence of bakeries, restaurants, traffic, waste areas, or bars, can help explain the sources of different smells in the city. For example, areas with heavy traffic may have stronger pollution odors, while food-related businesses may create pleasant smells during opening hours.
This type of data can be collected using publicly available sources, such as Google Maps or other, by linking the reported smells to the nearby activities or facilities.

NOTES: All of these datasets could face challenges related to *Volume*, since the amount of information required can grow very large, especially if collecting precise location and temporal data for many participants across the city. Among all the data types, of smell and location are strictly essential for creating a city smell map, while the other information such as sensation, meteorological conditions, time, and human activities is supplemental. These additional datasets are very useful for gaining a clearer and more detailed understanding of how smells vary across the urban environment.
However, all of this data can also raise concerns about *Value*, because gathering, collecting, and managing it may require significant resources and costs. There can also be potential issues related to privacy, especially when precise location information is recorded, as well as other logistical or ethical considerations when handling personal or sensitive data.

References

[1] Murray Parker, Dirk H. R. Spennemann, and Jennifer Bond. Methodologies for smellwalks and scentwalks—a critical review. *Chemical Senses*, 49:bjae005, 2024.

[2] CREATE Lab, Carnegie Mellon University. Smell mycity. https://smellmycity.org, 2025. Accessed: 2025-10-09.

[3] C. E. Campenni, E. J. Crawley, and M. E. Meier. Role of suggestion in odor-induced mood change. *Psychological Reports*, 94:1127–1136, 2004.

[4] M. A. Carskadon, J. M. Saletin, E. Van Reen, A. Bartz, C. Hart, H. Raynor, and R. S. Herz. Circadian influences on smell and taste detection thresholds: Preliminary results from adolescents. *Sleep*, 38:213–228, 2015.

Category	Description	Associated "V's" (Big Data Characteristics)	Why It's Important	Strategy of Collection
Smell	Free-text description from participants with 3 words max	*Variety, Veracity*	Allows natural, subjective input. Applying a text-mining or keyword clustering can group smells into themes (food, natural, pollution, and others)	Surveys, Apps or technological devices
Location	Street name, coordinates, or selected area (street, route, zone, city)	*Volume, Velocity, Variety*	Enables spatial mapping of smells and is essential for cartography and can help to reveal smell hotspots or patterns, such as: industrial or residential areas	Keeping track of coordinates, urban area or cell-phone localization
Sensation	Participant's evaluation of the smell: pleasant / unpleasant / neutral	*Veracity, Variability*	Adds emotional and qualitative dimension for understanding livability and comfort perception. Could be also useful for clustering and correlation analysis	Survey with some predefined answers and categories
Meteorological conditions (Meteo)	Weather at the moment of observation (temperature, wind, rain, humidity, etc.)	*Variability, Velocity*	Weather influences smell perception and intensity and including it can help normalize data and interpret temporal variation	Meteo database based on the location of smell data collection and time
Time	Exact time and date of the observation	*Velocity, Variability*	Allows temporal analysis (morning/evening, weekday/weekend, seasonal patterns). For example, "flower" scent more common in spring	Noting time when the subject express an opinion, can be hand made or via App
Human activities in the zone	Presence of possible smell-related sources (bakeries, traffic, waste areas, restaurants, etc.)	*Variety, Veracity, Volume*	Provides a contextual layer explaining smell sources and can help to identify correlations between human activity and smell profiles.	Using Google Maps or other services like this, also on site visits

Table 1: Data categories for the urban smell cartography, respective "V's" and collection strategies